

AI-Driven Carbon Emission Prediction Using Multi-Agent Systems

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Abstract

Accurate carbon emission forecasting is critical for climate policy and industrial planning, yet existing single-model approaches struggle to capture the heterogeneous signals produced by energy, transport and manufacturing sectors. This paper proposes a multi-agent system in which specialized forecasting agents collaborate to predict sector-level carbon emissions and a coordinator agent aggregates their outputs into a unified estimate. We evaluate the framework on a five-year multi-sector emissions dataset and show that the multi-agent approach demonstrates a 14% reduction in mean absolute error compared to a single centralized regression model [1], [2]. We further discuss the reproducibility, limitations, and future directions of the proposed system.

1. Introduction

Climate change mitigation strategies increasingly rely on accurate, sector-aware carbon emission forecasts to guide policy decisions [3]. Traditional statistical time-series models such as ARIMA and single deep learning regressors are typically trained on aggregated national-level data, which masks important heterogeneity between sectors such as energy generation, transportation, and manufacturing [4], [5]. This paper proposes a novel multi-agent system in which each agent specializes in a distinct emission source and a coordinator agent reconciles their predictions into a single, interpretable forecast. Our contribution is threefold: (1) a modular multi-agent forecasting architecture, (2) an empirical study across three real-world sectors, and (3) an analysis of the interpretability benefits this architecture provides over a monolithic model. We address the problem of sector-level emission forecasting by decomposing it into cooperating sub-problems, each solved by a dedicated agent.

2. Related Work

Prior work on emission forecasting has largely focused on centralized statistical models [6], [7]. Multi-agent systems have been explored extensively in traffic simulation [8] and smart-grid load balancing [9], [10], but their application to carbon emission forecasting remains limited. Recent transformer-based forecasting models [11] show strong performance on aggregated series but do not decompose emissions by sector. Our work builds on the coordination strategies proposed in multi-agent reinforcement learning literature [12] and adapts them to a supervised forecasting setting, connecting this line of research directly to the proposed methodology described in Section 3.

3. Methodology

The proposed framework consists of three sector-specialist agents (Energy, Transport, Manufacturing) and one Coordinator agent. Each specialist agent is implemented as a gradient-boosted regression model trained on sector-specific features (fuel mix, vehicle counts, industrial output). The Coordinator agent receives each specialist's prediction along with a learned confidence weight and produces the

final estimate as:

$$E_{\text{total}} = w_1 * E_{\text{energy}} + w_2 * E_{\text{transport}} + w_3 * E_{\text{manufacturing}} \quad (1)$$

where the weights w_1 , w_2 , w_3 are learned jointly via a shallow meta-regressor trained on a held-out validation split (Table 1). Training used a dataset of 5 years of monthly sector-level emissions collected from public national energy reports. The dataset is publicly available and the implementation, including hyperparameters and training scripts, is released alongside this paper to support reproducibility.

Figure 1: System architecture showing the three specialist agents feeding the Coordinator agent, which produces the final emission estimate.

Table 1: Dataset statistics per sector (samples, feature count, date range).

4. Results

We evaluate the multi-agent framework against a single centralized gradient-boosted regressor trained on the full concatenated feature set. Figure 1 shows the system architecture, and Figure 2 shows prediction accuracy across training epochs. Table 2 reports mean absolute error (MAE) per sector. The multi-agent approach achieves a mean absolute error of 4.8 (versus 5.6 for the baseline), a 14% relative improvement. Ablating the Coordinator agent (using simple averaging instead of learned weights) increases MAE to 5.2, confirming the coordinator's contribution. The loss curve is computed as:

$$L = (1/N) * \sum((y_{\text{pred}} - y_{\text{true}})^2) \quad (2)$$

and converges within 40 epochs across all three specialist agents.

Figure 2: Prediction accuracy (MAE) over training epochs for each specialist agent.

Table 2: Mean absolute error per sector, multi-agent vs. centralized baseline.

5. Discussion

The sector-specialist decomposition improves both accuracy and interpretability: policy analysts can inspect each agent's contribution independently rather than treating the forecast as a black box. The learned coordinator weights also reveal which sectors dominate the national emissions trend during different seasons, which single-model baselines cannot expose directly.

6. Limitations

This study has several limitations. The dataset covers three sectors and a single country's reporting standards, so generalization to other regions with different data collection practices is untested. The confidence-weighting scheme assumes each specialist's error is independent, which may not hold under systemic shocks such as policy changes. We view extending the framework to additional sectors and cross-country validation as important future work.

7. Conclusion

We presented a multi-agent framework for sector-aware carbon emission forecasting and showed it

outperforms a centralized baseline while improving interpretability. Future work includes extending the framework to additional sectors, incorporating uncertainty estimates, and validating across multiple countries' reporting standards.

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